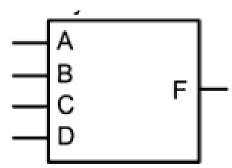
دانشکده مهندسی- گروه برق

عنوان درس: طراحی سیستمهای دیجیتال	**استاد درس: دکتر صیفی کاویان**
مدت امتحان: 100 دقیقه	**تاریخ امتحان: ١٤٠٥/٠٣/ ١٦**
استفاده از جزوه یا کتاب مجاز نیست.	

1) Design a Verilog model for **F=A' (B+CD')**. Use continuous assignment and logical operators A,B,C,D. **(5 P.)**

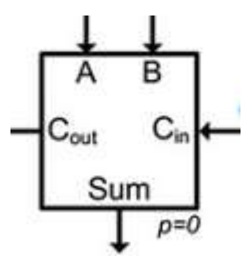
2) Design a Verilog model to implement the 4-input **Minterm List (2, 3, 9,11,15)** circuit. Use continuous assignment and logical operators. **(8 P.)**



3) Design a Verilog model for the **8-to-3 one-hot encoder**. Use continuous assignment and logical operators. **(10 P.)**

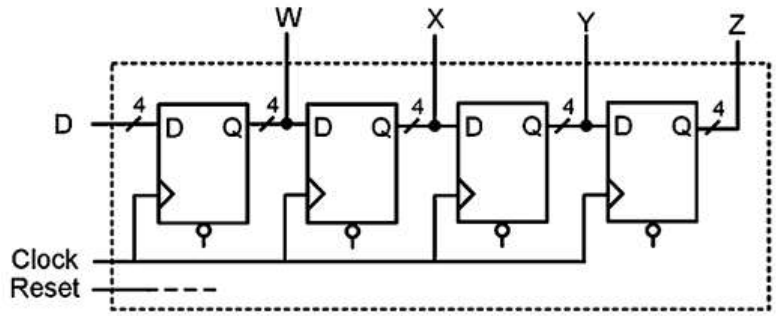
4)

A) Design a **behavioral** Verilog model for a FULL ADDER. **(5 P.)**

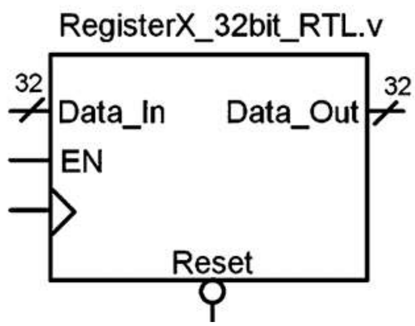


B) Use the designed FULL ADDER in a Verilog model to design an 8-bit ADDER. **(10 P.)**

5) Design a Verilog model for the **4 bit shift register**. The shift register has asynchronous reset. **(10 P.)**

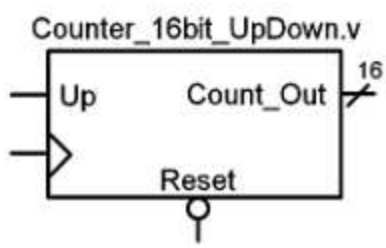


6) Design a Verilog model of a 32-bit register. The register has asynchronous reset and enable. **(10 P.)**



7) Design a Verilog model for a string detector circuit that takes as input a serial bit stream and outputs a '1' whenever the sequence ''01101'' occurs. How many Flip Flop is required? **(12 P.)**

8) Design a Verilog model for a 16-bit, binary up/down counter using a single procedural block. When Up is 1, the counter will increment. When Up is 0, the counter will decrement. The register has asynchronous reset **(10 P.)**



9) Design a Verilog model for the 16X8, synchronous, read/write memory system. **(10 P.)**